

Discrete Time Fourier Transform (DTFT) ①

⇒ If input signal is discrete & aperiodic ~~then~~

Analysis Equation

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

Synthesis Equation:-

$$x(n) = \frac{1}{2\pi} \int_{-\pi}^{\pi} X(e^{j\omega}) e^{j\omega n} d\omega$$

Fundamental
Equations
for DTFT
& IDTFT

$$x(n) \xleftrightarrow{\text{DTFT}} X(e^{j\omega})$$

$$x\left(\frac{1}{T}\right) \xleftrightarrow{\text{CTFT}} X(j\omega)$$

$X(e^{j\omega}) \rightarrow$ Spectrum of $x(n)$.

It is periodic with period 2π .

How ???

$$X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n}$$

$$X(e^{j(\omega+2\pi k)}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j(\omega+2\pi k)n}$$

$$= \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \cdot e^{-j2\pi kn} \quad \text{--- ①}$$

(2)

$$e^{-j(2\pi k)n} = \cos 2\pi kn - j \sin 2\pi kn$$

We know $\sin \pi, 2\pi, 3\pi, \dots$

$$\therefore \sin 2\pi kn = 0, \cos 2\pi kn = 1$$

$$e^{-j2\pi kn} = 1$$

Eq (1) will become ;

~~$$X(e^{j(\omega + 2\pi k)}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j(\omega + 2\pi k)n}$$~~

$$X(e^{j(\omega + 2\pi k)}) = \sum_{n=-\infty}^{\infty} x(n) e^{-j\omega n} \quad (1)$$

$$X(e^{j(\omega + 2\pi k)}) = X(e^{j\omega}) \quad \text{proved.}$$

$\therefore x(n)$ is discrete and aperiodic but its Spectrum $X(e^{j\omega})$ is periodic and continuous in nature.

-: Properties of DTFT :-

(3)

Similar to properties of CTFT

(i) Linearity :-

$$\text{If } \begin{aligned} x(n) &\xleftrightarrow{\text{DTFT}} X(e^{j\omega}) \\ y(n) &\xleftrightarrow{\text{DTFT}} Y(e^{j\omega}) \end{aligned}$$

$$\text{If } z(n) = ax(n) + by(n) \xleftrightarrow{\text{DTFT}} Z(e^{j\omega}) = aX(e^{j\omega}) + bY(e^{j\omega})$$

weighted sum of inputs

weighted sum of outputs

(ii) Time Shift :-

$$\begin{aligned} x(n) &\xleftrightarrow{\text{DTFT}} X(e^{j\omega}) \\ x(n-n_0) &\xleftrightarrow{\text{DTFT}} e^{-j\omega n_0} X(e^{j\omega}) \end{aligned}$$

(iii) Frequency Shift :-

$$\begin{aligned} x(n) &\xleftrightarrow{\text{DTFT}} X(e^{j\omega}) \\ e^{j\omega_0 n} x(n) &\xleftrightarrow{\text{DTFT}} X(e^{j(\omega-\omega_0)}) \end{aligned}$$

(iv) Scaling :-

$$\begin{aligned} x(n) &\xleftrightarrow{\text{DTFT}} X(e^{j\omega}) \\ x(np) &\xleftrightarrow{\text{DTFT}} X(e^{j(\frac{\omega}{p})}) \end{aligned}$$

5. Differentiation in Freq. Domain :-

(4)

$$x(n) \xleftrightarrow{\text{DTFT}} X(e^{j\omega})$$

$$-jn x(n) \xleftrightarrow{\text{DTFT}} \frac{d}{d\omega} X(e^{j\omega})$$

6. Convolution :-

$$x(n) \longleftrightarrow X(e^{j\omega})$$

$$y(n) \longleftrightarrow Y(e^{j\omega})$$

$$x(n) * y(n) \longleftrightarrow X(e^{j\omega}) Y(e^{j\omega})$$

7. Parseval's Theorem :-

$$\sum_{n=-\infty}^{\infty} |x(n)|^2 = \frac{1}{2\pi} \int_{-\pi}^{\pi} |X(e^{j\omega})|^2 d\omega$$

8. Time Reversal :-

$$x(n) \longleftrightarrow X(e^{j\omega})$$

$$x(-n) \longleftrightarrow X(e^{-j\omega})$$

Time reversal in
input

then corresponding reversal
in Freq. domain
as well.

DTFT Representation :-

(5)

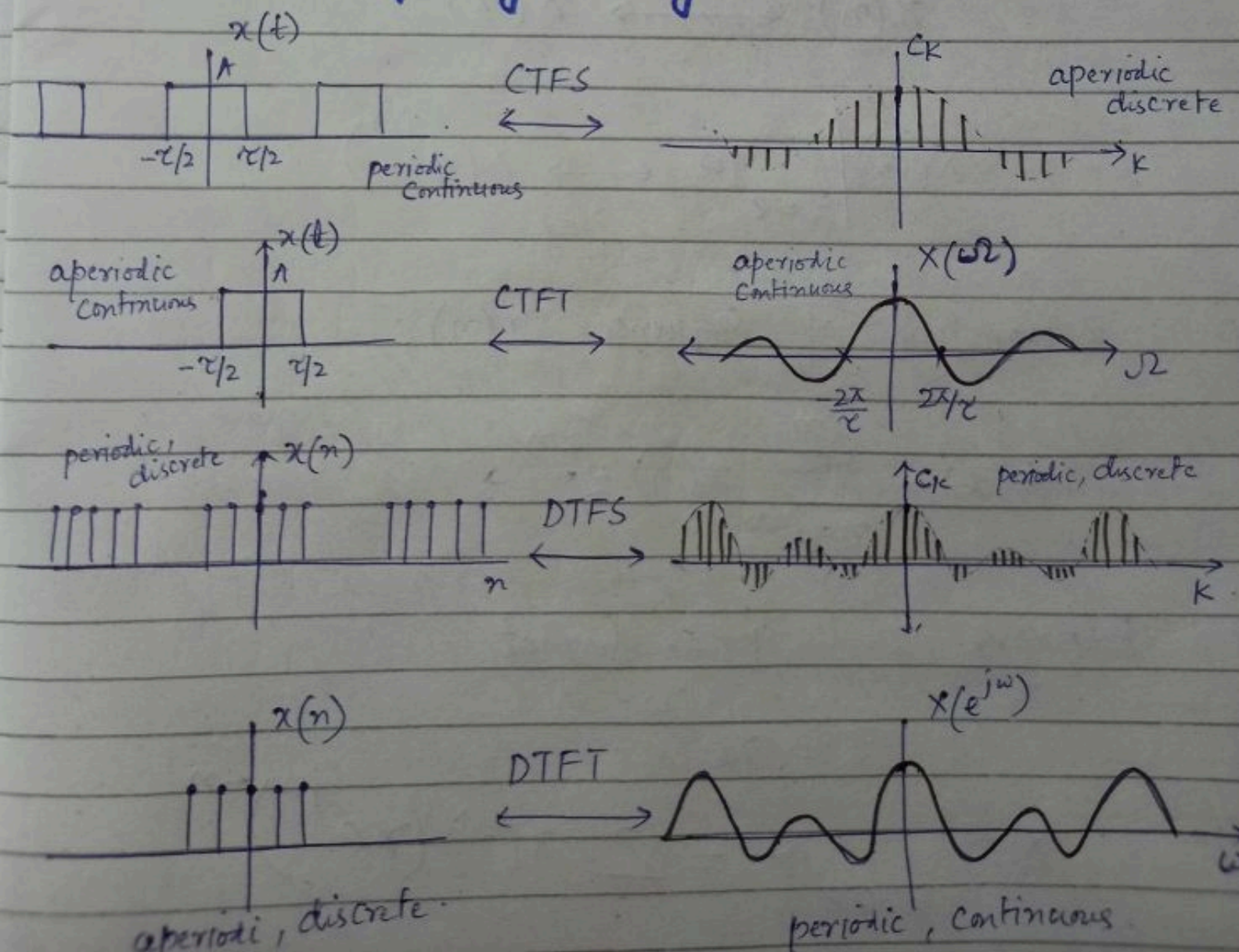
⇒ Magnitude Spectrum $|X(e^{j\omega})|$

⇒ Phase Spectrum $\angle X(e^{j\omega})$

⇒ Energy Spectrum $|X(e^{j\omega})|^2 = S_{xx}(e^{j\omega})$

aperiodic signals are energy signals
ie finite energy and zero power.

-: Frequency Analysis :-



- | | | |
|-------------------------|---------------|-----------------------|
| ① periodic, Continuous | <u>CTFS</u> → | aperiodic, discrete |
| ② aperiodic, Continuous | <u>CTFT</u> → | aperiodic, Continuous |
| ③ periodic, discrete | <u>DTFS</u> → | periodic, discrete |
| ④ aperiodic, discrete | <u>DTFT</u> → | aperiodic, Continuous |

⇒ Relationship of DTFT with Z-Transform :-

$$x(n) \xleftrightarrow{ZT} X(z)$$

$$X(z) \Big|_{z=e^{j\omega}} \longleftrightarrow X(e^{j\omega})$$

Z-Transform of sequence $x(n)$:

$$X(z) = \sum_{n=-\infty}^{\infty} x(n) z^{-n}$$

Express z in polar format

$$z = r e^{j\omega}$$

$$X(z) \Big|_{z=r e^{j\omega}} = \sum x(n) (r e^{j\omega})^{-n}$$

$$X(z) \Big|_{z=r e^{j\omega}} = \sum (x(n) r^{-n}) e^{-j\omega n} \quad (7)$$

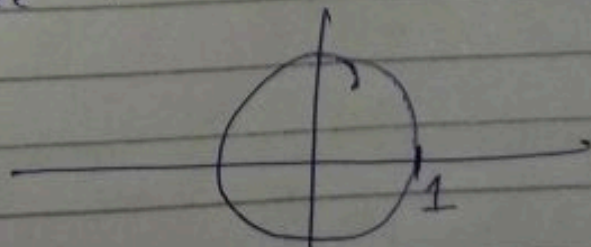
If $r = 1$

$$X(z) \Big|_{z=e^{j\omega}} = \sum x(n) e^{-j\omega n}$$

$\therefore$
 $\Rightarrow$ FT is viewed as Z-Transform of the sequence evaluated at unit circle

$\Rightarrow$ Z-Transform might exist anywhere in the z-plane but DTFT exists only at the unit circle i.e. $r=1$.

$\Rightarrow \therefore$ DTFT is the Z-Transform at unit circle.



$\therefore$ ROC should include unit circle only then DTFT = Z-Transform.